

Category	Criteria	Level 1 (Basic)	Level 2 (Intermediate)	Level 3 (Advanced)
Findable	Preservation or archival of data and software	Not archived or hosted in unreliable locations.	Archived data or tool in a trusted repository.	Both tool and data are archived in trusted, well-maintained repositories (e.g., Zenodo, ProteomeXchange).
	Version control	No version control, tool is deprecated or unreliable.	Minimal or inconsistent version control, tool still functional but not actively maintained.	Comprehensive version control with regular updates, clear changelogs, and active maintenance.
Accessible	Licensing and legal	Proprietary or restricted access requires payment or a limited free trial.	Open source but with restrictions (e.g., limited commercial use or missing code access).	Fully open-source with a permissive license (e.g., MIT, Apache), code publicly available.
	Community engagement	No evidence of community engagement or support.	Minimal engagement, limited to one platform (e.g., GitHub issues or a single forum).	Active engagement across multiple platforms (e.g., GitHub, forums, Q&A sites) with regular interactions.
Interoperable	Workflow management systems	Cannot integrate into workflow management systems.	Usable as part of a pipeline in workflow systems (e.g., Snakemake, Galaxy).	Fully compatible and integrated with workflow management systems for automation.
	Standardized input and output formats	Input/output formats are unknown or non-standard.	Formats are non-standard but can be converted with effort.	Uses standardized formats directly, ensuring seamless interoperability with other tools.
Reusable	Assessment against benchmarks or known standards	No benchmarking or testing reported.	Tested only at the time of initial tool publication.	Regular benchmarking and validation against widely accepted datasets with updates.
	Containerization and dependency management	Not containerized or dependency management is manual.	Partially containerized, e.g., as part of a broader pipeline or incomplete container.	Fully containerized (e.g., Docker, Singularity) with comprehensive dependency management.
	Manuals, documentation, logging, and reporting	Lacks a manual; minimal or unclear documentation, logging, or reporting.	Sufficient documentation but lacks a manual or comprehensive logging/reporting.	Detailed manual, thorough and clear documentation, and complete logging/reporting.
	Scalability and parallelization	The tool has unknown or no built-in scalability or parallelization capabilities. It struggles to handle large datasets efficiently and cannot leverage cloud or HPC resources.	The tool supports basic scalability. It can handle moderately large datasets but may not fully utilize advanced computational resources.	The tool is designed for high scalability and parallelization, capable of efficiently processing very large datasets. It supports distributed computing, cloud platforms, and HPC environments, with optimization for resource allocation and task distribution.